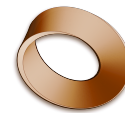


Name: _____

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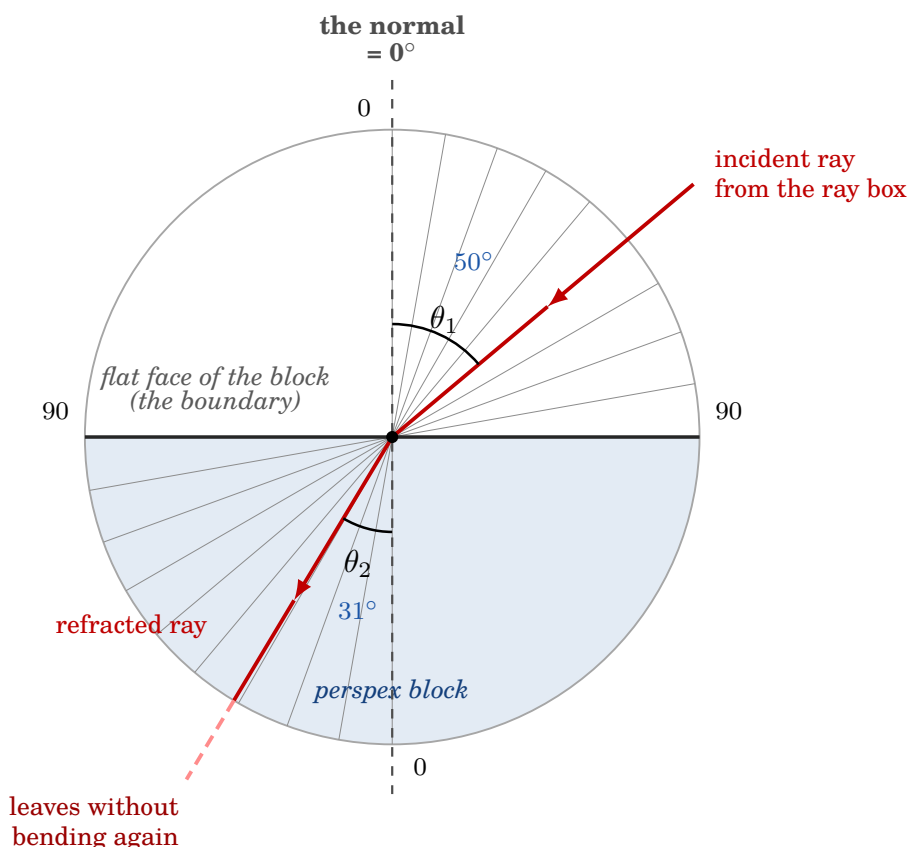
1.5 – Measuring Refraction

Unit 1: Light & EM Spectrum

Learning intentions

Last lesson we marched the boundary ourselves and found out **why** light bends. Today we find out **how much** it bends — and we learn the way of measuring it that you will need for your investigation.

Your block is a **semicircular** piece of perspex, and it sits on a printed card with **two protractors** on it. The block goes in the blue half, with its **flat face** along the line across the middle. You aim the ray at the **centre** of the card — and then you read both angles straight off the card. No tracing. No lifting.



Three things to notice on that diagram before you touch any apparatus.

- The ray bends when it crosses the **flat face** — that is the boundary between air and perspex. It bends **towards the normal**, because perspex slows light down.

- **The normal is the dashed vertical line, and on this card it is 0°.** Both angles are measured from it.

- On the way out, the ray leaves through the **curved** face — and it does **not** bend again. Question 3(e) asks you why.

1. The method

You have a **ray box**, a **semicircular perspex block**, and the **printed protractor card**. You will not need paper, a pencil trace, or your own protractor — the card is the protractor.

1. Put the block on the card in the **blue half**, curved side down. Line its **flat face** up exactly along the line across the middle, with the middle of the flat face on the **centre** of the card.
2. Switch the ray box on and make a **single** narrow ray. Aim it so that it hits the **centre** of the card — the point where the two protractors meet. Check this **every time**, before every reading.
3. Swing the ray box round until the incident ray lies along the 0° line of the top protractor. That is your first reading: an angle of incidence of 0° .
4. Read where the refracted ray comes out on the **bottom** protractor. Write it in the table.
5. Swing the ray box round to 10° , then 20° , and so on up to 80° , reading the bottom protractor each time.

The two instructions printed on your card

“Set up your ray box so that the light ray hits the **centre** of the protractors every time you take a reading.”

“Remember to measure from the **normal** (90° from the surface it is hitting), and take **the normal as 0°** .”

The one mistake that ruins the whole table. There are **two** scales printed on this card and only one of them is the right one. The right one reads **0° at the dashed normal** and 90° along the flat face. If your reading is coming out near 90° when the ray is nearly straight down, you are on the wrong scale — you have measured from the **surface** instead of from the **normal**. This is the same mistake that made pairs of angles add up to 90° in our reflection practical.

Safety. The ray box gets **hot**. Do not touch the bulb, and keep paper, sleeves and pencil cases out of it. If you smell burning, switch it off at the power supply and tell Mr. Rawson.

2. Your results

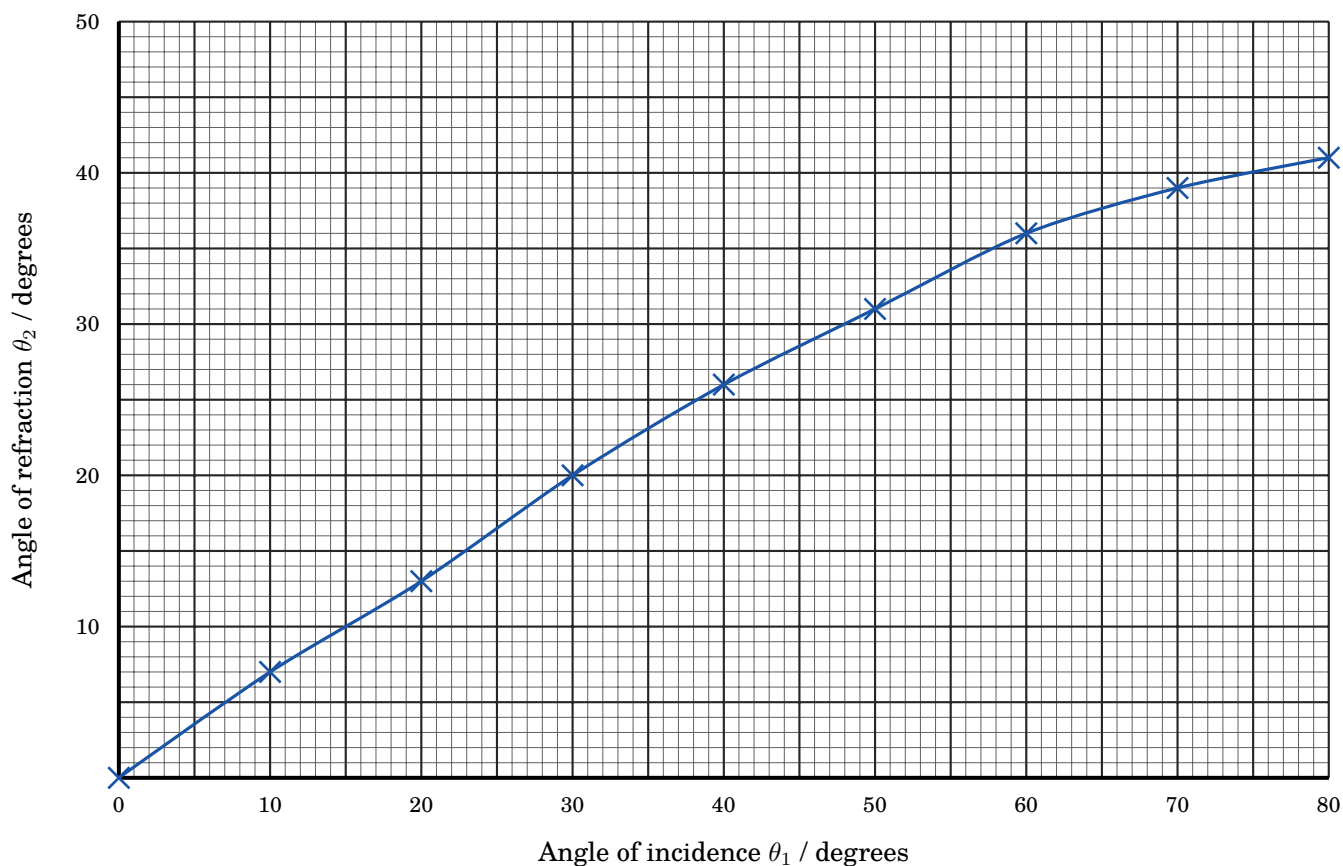
Read the angle of refraction to the **nearest degree**. The third column is just the first two subtracted — it tells you *how much* the ray bent.

Angle of incidence θ_1 , in air	Angle of refraction θ_2 , in perspex	How much it bent $\theta_1 - \theta_2$
0°	0°	0°
10°	$\approx 7^\circ$	3°
20°	$\approx 13^\circ$	7°
30°	$\approx 20^\circ$	10°
40°	$\approx 26^\circ$	14°
50°	$\approx 31^\circ$	19°
60°	$\approx 36^\circ$	24°
70°	$\approx 39^\circ$	31°
80°	$\approx 41^\circ$	39°

Values for perspex, $n = 1.49$. Anything within two or three degrees is a good measurement. What matters is that $\theta_2 < \theta_1$ every time, that the 0° row really does read 0°, and that the third column **grows** — the ray bends more and more as θ_1 increases.

3. Draw the graph

Plot **angle of refraction** (up) against **angle of incidence** (across). Plot your nine points with a sharp pencil, then draw a single smooth curve through them — not a zig-zag from dot to dot.



*The curve is the giveaway. It rises steeply at first and then **flattens off** — it is not a straight line, which is why dividing the angles will not give a constant in the extension.*

4. What your results show

(a) Compare each pair of angles. Is θ_2 always bigger than θ_1 , always smaller, or does it vary?

*θ_2 is always **smaller** than θ_1 . Going from air into perspex the ray always bends towards the normal.*

(b) Use the word **speed** to explain why.

*Light travels **slower** in perspex than in air. The side of the ray that reaches the perspex first slows down first, while the other side is still going quickly — so the ray swings round towards the normal. (This is the marching band crossing onto the long grass.)*

(c) What would happen to θ_2 if your block were made of **diamond** instead of perspex?

($n_{\text{perspex}} = 1.49$, $n_{\text{diamond}} = 2.42$)

*θ_2 would be even **smaller**. Diamond has a bigger n , so light is slower in it, so the ray bends more towards the normal.*

(d) Look at your 0° row. The ray came in **along the normal**. What did it do, and why?

It carried straight on without bending: $\theta_2 = 0^\circ$ as well. With $\theta_1 = 0^\circ$ no part of the ray reaches the perspex before any other part, so there is nothing to swing it round. It still slows down — it just does not change direction.

(e) The ray leaves the block through the **curved** face, and it does not bend there. Use the word **normal** to explain why not.

*Hint: the centre of the card is the centre of the circle, so the ray inside the block is travelling straight out along a **radius**.*

*The normal to a curved surface points straight out from the centre of the circle. The ray is travelling out along a radius, so it is already travelling **along the normal** — its angle of incidence at the curved face is 0° , and by part (d) a ray along the normal does not bend. That is the whole reason the block is this shape: it lets us study **one** boundary at a time.*

Extension

For each row of your table, work out $\theta_1 \div \theta_2$.

Then work out $\sin \theta_1 \div \sin \theta_2$.

One of those two calculations gives roughly the **same answer every time**. Which one — and what is that number close to?

*$\sin \theta_1 \div \sin \theta_2$ is the constant one, and it comes out at about **1.5** — the index of refraction of perspex. Dividing the angles themselves does not give a constant, which is exactly what the bend in your graph was telling you. (This is Snell's law, and you have just measured it.)*